

Propeller-Modulated Clutter Suppression Technique for UAV-borne Radar

Chenran Gao
Wuhan Radar Academy
Wuhan, China
13137187279@163.com

Wei Chen
Wuhan Radar Academy
Wuhan, China
chenwei1993x@163.com

Ming Hou
Wuhan Radar Academy
Wuhan, China
hm779911@163.com

Wenchong Xie*
Wuhan Radar Academy
Wuhan, China
xwch1978@163.com

Chengyin Liu
Wuhan Radar Academy
Wuhan, China
kworms@foxmail.com

Jiayi Zhang
Wuhan Radar Academy
Wuhan, China
18167909732@163.com

Yuanyi Xiong*
Wuhan Radar Academy
Wuhan, China
xyyafewa@163.com

Buqiu Tian
Wuhan Radar Academy
Wuhan, China
tbq_uestc@163.com

Abstract—When a UAV early-warning aircraft employs a turboprop engine, the high-speed rotation of the propellers induces parasitic frequency modulation in radar echoes. This effect shifts the mainlobe clutter and broadens the clutter region, significantly decreasing the radar’s moving-target detection performance. Existing methods for suppressing propeller-modulated clutter are either difficult to implement in practical engineering or demonstrate suboptimal effectiveness. To address this challenge, this paper proposes a propeller modulation clutter suppression method based on image denoising and sub-coherent processing interval three-dimensional space-time adaptive processing (sub-CPI 3D-STAP). The method is designed with engineering feasibility in mind and effectively enhances suppression performance solely through algorithmic-level improvements, thereby improving airborne radar detection capability. The performance of the proposed method is validated using both simulated and measured data.

Keywords—UAV-borne radar, Propeller-Modulated Clutter, 3D-STAP, Sub-CPI, Image Denoising

I. INTRODUCTION

Compared with conventional manned aircraft, unmanned aerial vehicles (UAVs) have emerged as the preferred platform for next-generation early-warning systems owing to their compact size, lightweight design, low maintenance requirements, and suitability for large-scale deployment, attracting global attention [1-5]. However, UAV early-warning platforms typically employ turboprop engines, whose high-speed propeller rotation inevitably modulates radar echo signals aligned with the propeller orientation. When the propeller lies within the pitch mainlobe, the clutter spectral distribution becomes particularly complex, significantly degrading moving-target detection performance. Consequently, the suppression of propeller modulation clutter represents a critical technical challenge that should be addressed in modern airborne radar systems.

Some studies have shown that the parasitic frequency modulation caused by propeller rotation widens the clutter region of airborne radar and degrades its moving target detection capability [6-12]. To suppress this type of clutter, experts and scholars have conducted a series of studies. R. Daba *et al.* [13] reduced clutter amplitude by optimizing propeller blade structural parameters, materials, and radar

antenna polarization characteristics. M. Ibrahim *et al.* [14] proposed an equalization method based on continuous reference signals to reduce modulation effects on RF signals. However, these methods often rely on propeller modulation models constructed from single-pulse signals, considering only unmodulated target signals, propeller-modulated target signals, and noise. In operational environments, airborne radar receives signals containing substantial clutter in addition to target echoes, making it extremely challenging to separate target signals from clutter for independent processing. Consequently, such methods are unsuitable for practical airborne radar applications.

Therefore, Y. Liu *et al.* [15] addressed short-range propeller-modulated clutter suppression using a three-dimensional (3D) sliding window combined with orthogonal projection from the perspective of clutter suppression. This method employs a blocking matrix and 3D sliding window to generate training samples and suppress short-range, nonstationary modulated clutter through sub-aperture orthogonal projection. However, this method is limited to short-range clutter. When the UAV propeller is positioned at the aircraft's nose, propeller modulation can also affect long-range clutter. Moreover, inaccurate estimation of the number of dominant eigenvalues can significantly degrade clutter suppression performance.

In summary, the existing propeller-modulated clutter suppression methods have the problem of being difficult to implement in engineering or achieving poor clutter suppression performance. For example, adjusting radar polarization characteristics increases hardware complexity and cost. Furthermore, existing methods fail to suppress long-range, nonhomogeneous clutter affected by propeller modulation, which remains a critical gap in practical applications.

To address these challenges, a propeller modulation clutter suppression method based on image denoising and sub-coherent processing interval three-dimensional space-time adaptive processing (sub-CPI 3D-STAP) is proposed, utilizing two key mechanisms. First, by exploiting the slow-time variation characteristics of propeller-modulated clutter, signal-to-clutter-plus-noise ratio (SCNR) can be substantially enhanced using image denoising techniques, without altering existing airborne radar system parameters, demonstrating strong engineering feasibility. Here, *slow-time* refers to the temporal sampling across successive pulse repetition

This work is supported in part by the National Natural Science Youth Fund of China under no. 62401622, Natural Science Foundation of Hubei Province under no. 2024AFB524, and Hubei Provincial Natural Science Foundation of China under no. 2026AFB765. (Corresponding authors: Yuanyi Xiong; Wenchong Xie.)

intervals. Second, by utilizing the elevation angle differences between the modulated clutter and target signals, an adaptive 3D notch in azimuth-elevation-Doppler space is established, which further improves clutter suppression performance.

II. CLUTTER MODEL

The geometric configuration corresponding to this modulation process is shown in Fig. 1. In this figure, O denotes the phase center of the radar array, O_1 denotes the geometric center of the aircraft propeller, P denotes the propeller scattering point, θ_c denotes the azimuth angle between the clutter scattering cell and the array axis, and φ_c denotes the elevation angle. For the l -th range gate, the direction vector of the i_c -th clutter scattering cell along the radar line of sight is expressed as

$$\mathbf{K}(\theta_{c,l}, \varphi_{c,l}) = (\cos \theta_{c,l} \cos \varphi_{c,l}, \sin \theta_{c,l} \cos \varphi_{c,l}, -\sin \varphi_{c,l}). \quad (1)$$

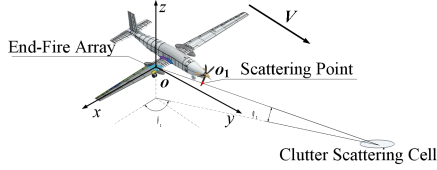


Fig. 1. Schematic of the geometric relationships among the airborne radar, propeller scattering points, and clutter scattering cells.

Based on the spatial relationship between the radar array and the propeller, a range gate is divided into three zones, M, U and S, as shown in Fig. 2. Clutter scattering cells in Zone M are subject to propeller modulation because their echo signal paths coincide with the propeller orientation, whereas those in Zone U remain unmodulated. The clutter scattering cells located in Zone S are affected by the obstruction of the fuselage, so the signals from this zone will not be received by the array.

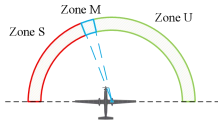


Fig. 2. Schematic Diagram of Propeller Modulation Zoning.

Therefore, the clutter signal vector received by the m -th array element in the l -th range gate is

$$\mathbf{X}_{c,ml} = \left(\sum_{n_r=1}^{N_r} \sum_{i_c=1}^{N_{cM,l}} a'_{i_c,l} a_{i_c,l} \hat{a}_{mi,l} \hat{\mathbf{S}}_t(\bar{\mathbf{f}}_{di,n_r}) + \sum_{n_r=1}^{N_r} \sum_{i_c=1}^{N_{cU,l}} a'_{i_c,l} a_{i_c,l} a_{mi,l} \mathbf{S}_t(\bar{\mathbf{f}}_{di,n_r}) \right) e^{j2\pi \frac{\mathbf{K}(\theta_{c,l}, \varphi_{c,l}) \cdot \mathbf{d}_m}{\lambda}}. \quad (2)$$

where $a'_{i_c,l}$ denotes the random fluctuation introduced by the clutter block reflection, $a_{i_c,l}$ denotes the statistical distribution characteristics of clutter amplitude, $a_{mi,l}$ denotes the unmodulated echo amplitude of clutter scattering cells received by the m -th array element, $\hat{a}_{mi,l}$ denotes the modulated echo amplitude of the clutter scattering cells received by the m -th array element, $N_{cM,l}$ denotes the number of clutter scattering cells located in Zone M within the l -th range gate, $N_{cU,l}$ denotes the number of clutter scattering cells located in Zone U within the l -th range gate, N_r denotes the number of range ambiguities, $\mathbf{S}_t(\bar{\mathbf{f}}_{di,n_r})$ denotes the clutter time-domain steering vector, $\hat{\mathbf{S}}_t(\bar{\mathbf{f}}_{di,n_r})$ denotes the clutter time-domain steering vector after propeller modulation, $\mathbf{d}_m$ denotes the position vector of the m -th array element with

respect to the array phase center, and λ denotes the wavelength.

When a target signal enters the array through the mainlobe, the corresponding airborne radar echo in the range gate where the target is located, including propeller-modulated clutter, can be expressed as

$$\mathbf{X}_{m,l} = a_{t,m} \mathbf{S}_t(\bar{\mathbf{f}}_{dt}) + \mathbf{X}_{c,ml} + \mathbf{n}, \quad (3)$$

where $a_{t,m}$ denotes amplitude of the target echo signal received by the m -th array element, $\mathbf{S}_t(\bar{\mathbf{f}}_{dt})$ denotes the target temporal steering vector, $\mathbf{n}$ denotes noise.

Subsequently, the signals received by all array elements of the receiving antenna are combined in column-major order to form the airborne radar echo signal corresponding to this range gate. It can be expressed as

$$\mathbf{X}_{tot,l} = [\mathbf{X}_{1,l}, \mathbf{X}_{2,l}, \dots, \mathbf{X}_{M,l}]^T, \quad l = 1, 2, \dots, L, \quad (4)$$

where M denotes the total number of array elements, and L denotes the total number of range gates.

To study the slow-time dimension effects on propeller-modulated clutter, pulses are sequentially removed to generate multiple range-Doppler spectra for analysis. As shown in Fig. 3, the clutter region modulated by the propellers exhibit a much higher variance than the mainlobe clutter, revealing strong distribution fluctuations and weak slow-time correlations for such clutter.

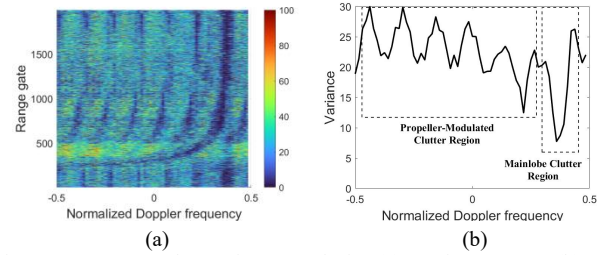


Fig. 3. Range-Doppler variance analysis. (a) Variance map; (b) Mean variance map.

III. PRINCIPLE OF THE PROPOSED METHOD

This section exploits the weak correlation of propeller-modulated clutter in the slow-time domain to propose a method for its suppression in airborne radar systems. The proposed method mainly consists of the subarray division, Sub-CPI 3D-STAP processing, and image denoising stages.

A. Subarray Division Stage

Owing to the angular separation between propeller-modulated clutter and target signals in the elevation dimension, 3D-STAP is employed for preliminary clutter suppression. This method employed 3D degrees of freedom across azimuth, elevation, and time to form an adaptive 3D notch.

The echo data corresponding to a given range gate, as received by the subarrays of the array, can be expressed as

$$\mathbf{X}_{s,l} = (\mathbf{T}_s \otimes \mathbf{I}_K)^H \mathbf{X}_{tot,l}, \quad l = 1, 2, \dots, L, \quad (5)$$

where

$$\mathbf{T}_s = \begin{bmatrix} \mathbf{G}_0 & \mathbf{0}_{Q_s \times 1} & \mathbf{0}_{Q_s \times 1} \\ & \mathbf{G}_0 & \ddots \\ \mathbf{0}_{(M-Q_s) \times 1} & \mathbf{0}_{(M-2Q_s) \times 1} & \mathbf{G}_0 \end{bmatrix}, \quad (6)$$

$$\mathbf{G}_0 = \left[1, e^{\frac{j2\pi d}{\lambda} \sin \theta_0 \cos \varphi_0}, \dots, e^{\frac{j(Q_s-1)2\pi d}{\lambda} \sin \theta_0 \cos \varphi_0} \right]^T, \quad (7)$$

Q_s denotes the number of elements in the array subarray, d denotes the spacing between elements in the subarray, θ_0 denotes the azimuth angle of the beam pointing, and φ_0 denotes the elevation angle of the beam pointing.

B. Sub-CPI 3D-STAP Processing

According to the mechanism of propeller modulation, the modulation effect exerted on each pulse is different. The random removal of a single pulse induces pronounced fluctuations in the propeller-modulated clutter region. Utilizing this property, the original pulse sequence is partitioned into multiple sub-CPI datasets, each of which is processed using 3D-STAP. In the resulting outputs, signals unaffected by propeller modulation exhibited stable power distributions, whereas signals affected by propeller modulation displayed significant variations in power distribution.

Using this principle, sub-CPI datasets are generated by successively removing individual pulses. When the k -th pulse ($k = 1, 2, \dots, K$) is removed, the echo data vector for the k -th sub-CPI corresponding to the l -th range gate is expressed as

$$\mathbf{X}_{\text{sub-CPI},kl} = (\mathbf{I}_N \otimes \mathbf{P}_k) \mathbf{X}_{s,l}, \quad (8)$$

where $\mathbf{P}_k$ denotes the sub-CPI selection matrix of dimension $K \times K$, and the expression is

$$\mathbf{P}_k = \begin{bmatrix} \mathbf{I}_{k-1} & \mathbf{0}_{(k-1) \times 1} & \mathbf{0}_{(k-1) \times (K-k)} \\ \mathbf{0}_{1 \times (k-1)} & 0 & \mathbf{0}_{1 \times (K-k)} \\ \mathbf{0}_{(K-k) \times (k-1)} & \mathbf{0}_{(K-k) \times 1} & \mathbf{I}_{K-k} \end{bmatrix}. \quad (9)$$

After estimating the clutter-plus-noise covariance matrix using selected training samples, each sub-CPI dataset is processed through 3D-STAP to generate a set of range-Doppler spectrum maps. During 3D-STAP processing, the dimension reduction matrix for the j -th Doppler channel is therefore defined as

$$\mathbf{T}_j = \mathbf{I}_N \otimes \tilde{\mathbf{F}}_j, \quad (10)$$

As the proposed method employs three adjacent Doppler channels for adaptive processing, the Doppler dimension reduction matrix can be expressed as

$$\tilde{\mathbf{F}}_j = [\mathbf{F}_{j-1}, \mathbf{F}_j, \mathbf{F}_{j+1}]. \quad (11)$$

The space-time adaptive weight vector for the j -th Doppler channel under test in the l -th range gate is then defined as

$$\tilde{\mathbf{W}}_j = \frac{\tilde{\mathbf{R}}_j^{-1} \tilde{\mathbf{S}}_j}{\tilde{\mathbf{S}}_j^H \tilde{\mathbf{R}}_j^{-1} \tilde{\mathbf{S}}_j}, \quad (12)$$

where

$$\tilde{\mathbf{R}}_j = \mathbf{T}_j^H \mathbf{R}_{c,l} \mathbf{T}_j, \quad (13)$$

$$\mathbf{R}_{c,l} = E(\mathbf{X}_{s,l} \mathbf{X}_{s,l}^H), \quad (14)$$

$$\tilde{\mathbf{S}}_j = \mathbf{T}_j^H \mathbf{S}_j, \quad (15)$$

$\mathbf{S}_j$ denotes the space-time steering vector of the j -th Doppler channel, $\tilde{\mathbf{F}}_j$ denotes the Doppler filtering weight vector corresponding to the j -th Doppler channel to be detected, and $\mathbf{R}_{c,l}$ is estimated from selected training samples. The detailed

selection procedure is described in [16] and is not reiterated here.

Once the space-time adaptive weight vector is obtained, the corresponding output signal is expressed as

$$y_{klj} = \tilde{\mathbf{W}}_j^H \mathbf{T}_j^H \mathbf{X}_{\text{sub-CPI},kl}, j = 1, 2, \dots, K, l = 1, 2, \dots, L. \quad (16)$$

Finally, the range-Doppler spectrum map, $\mathbf{Y}_k$, is constructed from the output signal and expressed as

$$\mathbf{Y}_k = \begin{bmatrix} y_{k11} & y_{k12} & \dots & y_{k1K} \\ y_{k21} & y_{k22} & \dots & y_{k2K} \\ \vdots & \vdots & \dots & \vdots \\ y_{kL1} & y_{kL2} & \dots & y_{kLK} \end{bmatrix}. \quad (17)$$

By removing pulses at different positions and repeating this procedure, K sets of output data are generated, forming the set $\mathbf{Y}_0 = \{\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_K\}$.

C. Image Denoising Processing

As each range-Doppler spectrum map in the output set $\mathbf{Y}_0$ is generated using the same number of pulses, the Doppler resolution remains constant. Signals unaffected by propeller modulation exhibit stable power distributions, resulting in fixed channel positions within the maps. In contrast, the power distribution of propeller-modulated clutter fluctuates with the positions of the removed pulses, causing the corresponding channel locations to vary randomly across the map set. This behavior renders propeller-modulated clutter analogous to noise points in image processing.

Exploiting this property, propeller-modulated clutter is further suppressed via spectral map denoising. The procedure involves performing weighted averaging along the y -axis of the 3D map. To preserve the target signal power, the sum of elements in the weighting vector is constrained to a fixed value. In the j -th Doppler channel, the weighted output is required to satisfy the following equation:

$$\begin{cases} \min_{\mathbf{W}_{\text{image},j}} \sum_{l=1}^L |\mathbf{W}_{\text{image},j}^H \mathbf{Y}_{s,lj}|^2 \\ \text{s.t.} \sum_{k=1}^K w_{\text{image},kj} = 1 \end{cases}, \quad (18)$$

where

$$\mathbf{W}_{\text{image},j} = [w_{\text{image},1j}, w_{\text{image},2j}, \dots, w_{\text{image},Kj}]^T, \quad (19)$$

$$\mathbf{Y}_{s,lj} = [y_{1lj}, y_{2lj}, \dots, y_{Klj}]^T. \quad (20)$$

By solving the aforementioned constrained optimization problem via the Lagrangian function, we can obtain:

$$\mathbf{W}_{\text{image},j} = \mu_j \hat{\mathbf{R}}_j^{-1} \mathbf{u}_0, \quad (21)$$

where

$$\mu_j = \frac{1}{\mathbf{u}_0^T \hat{\mathbf{R}}_j^{-1} \mathbf{u}_0}, \quad (22)$$

$$\hat{\mathbf{R}}_j = \sum_{l=1}^L \tilde{\mathbf{R}}_{lj}, \quad (23)$$

$$\mathbf{u}_0 = \underbrace{[1, 1, \dots, 1]^T}_K. \quad (24)$$

After obtaining the weighting coefficients, the final output expression is expressed as

$$z_{lj} = \mathbf{W}_{\text{image},j}^H \mathbf{Y}_{s,lj}, j = 1, 2, \dots, K, l = 1, 2, \dots, L. \quad (25)$$

IV. SIMULATION PARAMETER SETTING

A. Simulation Parameter Setting

In this simulation part, an end-fire array comprising 6 rows $\times$ 24 columns $\times$ 2 layers is employed. The array is divided into 48 subarrays, each measuring 6 rows $\times$ 1 column $\times$ 1 layer. The simulation parameters are summarized in Table I, with the elevation angle of the rotation center defined as the angular separation in the elevation dimension between the full-array phase center and the propeller rotation center.

TABLE I. SIMULATION PARAMETERS OF AIRBORNE RADAR.

Parameter Category	Name	Value
System Parameters	Wavelength	0.4 m
	Pulse repetition frequency	2000 Hz
	Number of pulses	64
Antenna Parameters	Row element spacing/m	0.2
	Column element spacing/m	0.1
	Array pointing	North
Aircraft Parameters	Altitude/km	8
	Flight direction	8° north by east
	Velocity	220 m/s
Propeller Parameters	Number of blades	5
	Root distance from rotation center/m	1
	Tip distance from rotation center/m	3
	Elevation angle of rotation center	15°
	Rotational speed	2400rpm

B. Suppression Effect of Propeller-Modulated Clutter

Fig. 4 compares the range-Doppler maps processed by 3D-STAP and the proposed method under the scenario of propeller-modulated clutter. Specifically, Figs. 4(a), 4(b), and 4(c) show the range-Doppler spectrum maps obtained by 3D-STAP, the uniform weight averaging method, and the proposed method, respectively, whereas Fig. 4(d) shows the clutter power distribution map normalized with the mainlobe clutter as the reference. The results demonstrate that in the propeller-modulated clutter region, the average propeller-modulated clutter power processed by the proposed method is 2.43 dB lower than that processed by 3D-STAP and 1.48 dB lower than that processed by the uniform weight averaging method.

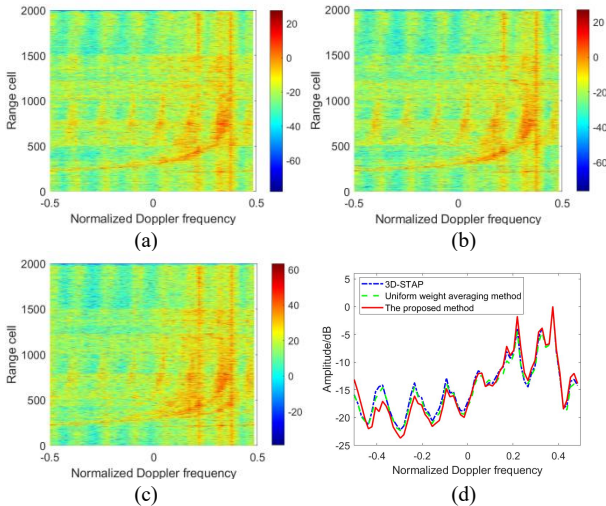


Fig. 4. Range-Doppler spectrum maps processed by different methods. (a) 3D-STAP; (b) Uniform weight averaging method; (c) Proposed method; (d) Power distribution of propeller-modulated clutter.

The suppression performance of propeller-modulated clutter is further evaluated by inserting target signals into Doppler channels occupied by modulated clutter and using the output signal power as a reference. Fig. 5 compares the power distributions in the target Doppler channels for both methods, with output power normalized to the target power. Compared to 3D-STAP, the proposed method enhanced the SCNR by approximately 3.58dB while substantially suppressing the propeller-modulated clutter.

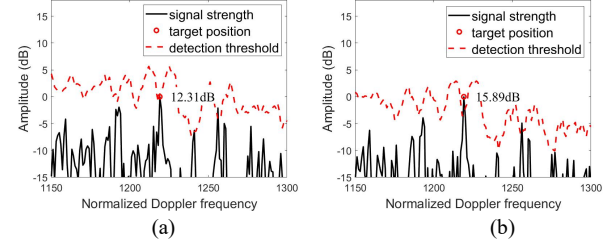


Fig. 5. Amplitude distribution of Doppler channels before and after suppression. (a) 3D-STAP; (b) Proposed method.

C. Processing Results of Measured Data

Fig. 6 compares the measured data processing results of the 3D-STAP and the proposed method under the condition of propeller-modulated clutter. The results show that the SCNR of the 3D-STAP is 23.96dB, while that of the proposed method reaches 26.77dB. The proposed method achieves SCNR improvement of approximately 2.81dB, demonstrating its enhanced capability to suppress propeller-modulated clutter.

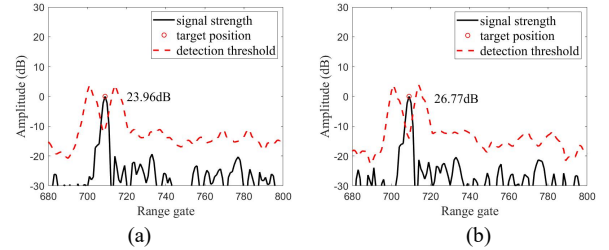


Fig. 6. Doppler channel power distribution for measured data. (a) 3D-STAP; (b) Proposed method.

V. CONCLUSION

This paper addresses the challenge of propeller-modulated clutter in UAV-borne radars, which significantly degrades moving target detection performance. It proposes a sub-CPI 3D-STAP suppression method enhanced with image denoising. The key contribution of this method lies in utilizing the weak correlation of propeller-modulated clutter in the slow-time dimension, along with the angular separation in elevation between the propeller-modulated clutter and target. Simulation and measured data demonstrate that, compared with conventional 3D-STAP, the proposed method achieved an improvement exceeding 2dB in the SCNR of targets affected by propeller-modulated clutter. For targets unaffected by modulated clutter, when the pulse number exceeds 200, the SNR deviation between the proposed method and the traditional PD processing remains below 0.2 dB. This indicates that the method has a negligible impact on the detection performance of airborne radar in clear regions.

REFERENCES

- [1] J. He, Z. Liao, Z. Qiu, and K. Duan, "Modeling and analysis for manned-unmanned bistatic airborne radar," 6th Intl Conf. on inf. commun. and Signal processing. (ICICSP), Xi'an, China, 2023, pp. 585-591. DOI: 10.1109/ICICSP59554.2023.10390839.
- [2] J. He, H. Wang, and K. Duan, "Cramér-Rao bound assessment for manned-unmanned bistatic airborne radar," IET Int. Radar Conf. (IRC 2023), Chongqing, China, 2023, pp. 1399-1406. DOI: 10.1049/icp.2024.1291.
- [3] J. Xiao, F. Chen, S. Cao, X. Hu, X. Cao, and W. Ren, "Modeling and simulation of manned and unmanned cooperative reconnaissance and surveillance scenario based on DoDAF," IEEE 19th Conf. on Ind. Electron. and Appl. (ICIEA), Kristiansand, Norway, 2024, pp. 1-5. DOI: 10.1109/ICIEA61579.2024.10664859.
- [4] H. N. Qureshi and A. Imran, "On the tradeoffs between coverage radius, altitude, and beamwidth for practical UAV deployments," IEEE Trans. Aerosp. Electron. Syst., vol. 55, no. 6, pp. 2805-2821, Dec. 2019. DOI: 10.1109/TAES.2019.2893082.
- [5] L. Chen, C. Wei, and H. Duan, "Multiple unmanned aerial vehicles coalition formation and control for collaborative defense mission," IEEE Trans. Aerosp. Electron. Syst., vol. 60, no. 5, pp. 6095-6109, Oct. 2024. DOI: 10.1109/TAES.2024.3402658.
- [6] J. Martin and B. Mulgrew, "Analysis of the theoretical radar return signal form aircraft propeller blades," IEEE Intl Conf. on Radar, Arlington, VA, USA 1990, 1990, pp. 569-572. DOI: 10.1109/RADAR.1990.201091.
- [7] C. Li, Y. Yang, Y. Li, X. Wang, and X. Huang, "Simulation and analysis of polarimetric signatures for propeller modulation," Intl Conf. on Radar Syst. (Radar 2017), Belfast, 2017, pp. 1-4. DOI: 10.1049/cp.2017.0534.
- [8] H. Krukovich and T. Pratt, "Frequency analysis of rotational motion using dual-polarized pulse-Doppler radar," IEEE Trans. Aerosp. Electron. Syst., vol. 58, no. 3, pp. 2611-2615, Jun. 2022. DOI: 10.1109/TAES.2021.3124204.
- [9] R. Chongji and D. Jianjiang, "The derivation of a universal model for the modulated characteristics of aircraft rotating parts," CIE Intl Conf. on Radar, Shanghai, China 2006, 2006, pp. 1-5. DOI: 10.1109/ICR.2006.343426.
- [10] B. Zhou, J. Li, H. Dai, D. Kong, and B. Jiao, "Analysis and simulation of modulation effect by aircraft rotating parts," IEEE Intl Conf. on Microw. and Millimeter Wave Technol. (ICMMT), Beijing 2016, 2016, pp. 467-469. DOI: 10.1109/icmmt.2016.7761810.
- [11] J. L. Garry and G. E. Smith, "Experimental observations of micro-Doppler signatures with passive radar" in IEEE Trans. Aerosp. Electron. Syst., vol. 55, no. 2, pp. 1045-1052, Apr. 2019. DOI: 10.1109/TAES.2019.2895584.
- [12] W. Li, Y. Xiong, C. Wei, P. Chen, D. Liang, and Z. Peng, "Near-Field Micro-Doppler Effect and Spectrum Characteristics of Rotating Targets: Theoretical Analysis and Parameter Estimation Method," IEEE Trans Instrument Meas., vol. 72, pp. 1-12, 2023, Art no. 8006212, DOI: 10.1109/TIM.2023.3322490.
- [13] R. Daba and B. Polat, "An analytical approach to rotor blade modulation," Wave Motion, vol. 105, no. 3, p. 102762, 2021. DOI: 10.1016/j.wavemoti.2021.102762.
- [14] M. Ibrahim and S. Ekin, "Propeller modulation equalization via reference tones," MILCOM IEEE Military Communications Conf. (MILCOM), Washington, DC, USA, pp. 1-5. DOI: 10.1109/MILCOM61039.2024.10774041.
- [15] Y. Liu, L. Ren, Z. Liu, H. Fan, and Q. Liu, "Propeller-modulation clutter suppression for airborne radar based on 3D sliding window and orthogonal projection," IEEE Intl Conf. on Signal, Information and Data Process. (ICSIDP), Zhuhai, China, 2024, pp. 1-5. DOI: 10.1109/ICSIDP62679.2024.10868995.
- [16] C. Gao, W. Xie, Y. Xiong, W. Chen and B. Tian, "A training sample selection method with fusing GIP statistic and geographic information for airborne radar," Electronics Letters, vol. 61, no. 1, 2025. DOI: 10.1049/ell2.70345.